

CO4-AT1 Industrial problem solving

CSA0904-Programming in Java

Kandath Haneesh 192324084

Inventory Management System

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import javax.swing.table.TableRowSorter;
import java.awt.*;
import java.awt.event.*;

public class InventoryManagement extends JFrame
implements ActionListener {

    // Form fields
    JTextField idField, nameField, priceField, quantityField,
searchField;

    // Buttons
    JButton addButton, updateButton, deleteButton,
clearButton;

    // Table
    JTable table;
    DefaultTableModel model;

    public InventoryManagement() {

        setTitle("Inventory Management System");
        setSize(800, 550);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        // ----- MENU BAR -----
```

```
JMenuBar menuBar = new JMenuBar();

JMenu menu = new JMenu("Product");

JMenuItem clearMenu = new JMenuItem("Clear");
JMenuItem exitMenu = new JMenuItem("Exit");

clearMenu.addActionListener(e -> clearFields());
exitMenu.addActionListener(e -> System.exit(0));

menu.add(clearMenu);
menu.add(exitMenu);

menuBar.add(menu);
setJMenuBar(menuBar);

// ----- FORM PANEL -----
JPanel formPanel = new JPanel(new GridLayout(5, 2,
10, 10));

formPanel.setBorder(BorderFactory.createTitledBorder("P
roduct Details"));

formPanel.add(new JLabel("Product ID:"));
idField = new JTextField();
formPanel.add(idField);

formPanel.add(new JLabel("Product Name:"));
nameField = new JTextField();
formPanel.add(nameField);

formPanel.add(new JLabel("Price:"));
priceField = new JTextField();
formPanel.add(priceField);

formPanel.add(new JLabel("Quantity:"));
```

```

quantityField = new JTextField();
formPanel.add(quantityField);

// Buttons
JPanel buttonPanel = new JPanel();

addButton = new JButton("Add");
updateButton = new JButton("Update");
deleteButton = new JButton("Delete");
clearButton = new JButton("Clear");

addButton.addActionListener(this);
updateButton.addActionListener(this);
deleteButton.addActionListener(this);
clearButton.addActionListener(this);

buttonPanel.add(addButton);
buttonPanel.add(updateButton);
buttonPanel.add(deleteButton);
buttonPanel.add(clearButton);

// ----- SEARCH PANEL -----
JPanel searchPanel = new JPanel(new
FlowLayout(FlowLayout.LEFT));

searchPanel.add(new JLabel("Search:"));

searchField = new JTextField(25);
searchPanel.add(searchField);

// ----- TABLE -----
String[] columns = {
    "Product ID",
    "Product Name",
    "Price",
    "Quantity"

```

```
};
```

```
model = new DefaultTableModel(columns, 0);
```

```
table = new JTable(model);
```

```
// Allows sorting by clicking table headers
```

```
TableRowSorter<DefaultTableModel> sorter =  
    new TableRowSorter<>(model);
```

```
table.setRowSorter(sorter);
```

```
// Search/filter functionality
```

```
searchField.getDocument().addDocumentListener(  
    new javax.swing.event.DocumentListener() {
```

```
        public void insertUpdate(  
            javax.swing.event.DocumentEvent e) {  
            filter();  
        }  
    }
```

```
        public void removeUpdate(  
            javax.swing.event.DocumentEvent e) {  
            filter();  
        }  
    }
```

```
        public void changedUpdate(  
            javax.swing.event.DocumentEvent e) {  
            filter();  
        }  
    }
```

```
    private void filter() {
```

```
        String text = searchField.getText();
```

```
        if (text.trim().length() == 0) {
```

```

        sorter.setRowFilter(null);
    } else {
        sorter.setRowFilter(
            RowFilter.regexFilter(
                "(?i)" + text
            )
        );
    }
}
);

```

```

// Click table row to load data into form
table.addMouseListener(new MouseAdapter() {

```

```

    public void mouseClicked(MouseEvent e) {

```

```

        int row = table.getSelectedRow();

```

```

        if (row >= 0) {

```

```

            int modelRow =
                table.convertRowIndexToModel(row);

```

```

            idField.setText(
                model.getValueAt(modelRow,
0).toString());

```

```

            nameField.setText(
                model.getValueAt(modelRow,
1).toString());

```

```

            priceField.setText(
                model.getValueAt(modelRow,
2).toString());

```

```

        quantityField.setText(
            model.getValueAt(modelRow,
3).toString());
    }
}
});

// ----- MAIN PANEL -----
JPanel topPanel = new JPanel(new BorderLayout());

topPanel.add(formPanel, BorderLayout.CENTER);
topPanel.add(buttonPanel, BorderLayout.SOUTH);

add(topPanel, BorderLayout.NORTH);
add(searchPanel, BorderLayout.CENTER);
add(new JScrollPane(table), BorderLayout.SOUTH);

// Better table layout
JPanel mainPanel = new JPanel(new BorderLayout());

mainPanel.add(topPanel, BorderLayout.NORTH);
mainPanel.add(searchPanel, BorderLayout.CENTER);
mainPanel.add(new JScrollPane(table),
BorderLayout.SOUTH);

add(mainPanel);

// Sample products
model.addRow(new Object[]{"P101", "Laptop",
55000, 10});
model.addRow(new Object[]{"P102", "Mouse", 800,
25});
model.addRow(new Object[]{"P103", "Keyboard",
1500, 15});

setVisible(true);

```

```

}

// ----- BUTTON EVENTS -----

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == addButton) {
        addProduct();
    }

    else if (e.getSource() == updateButton) {
        updateProduct();
    }

    else if (e.getSource() == deleteButton) {
        deleteProduct();
    }

    else if (e.getSource() == clearButton) {
        clearFields();
    }
}

// ----- ADD -----

void addProduct() {

    if (idField.getText().isEmpty() ||
        nameField.getText().isEmpty() ||
        priceField.getText().isEmpty() ||
        quantityField.getText().isEmpty()) {

        JOptionPane.showMessageDialog(
            this,
            "Please enter all details."
        );
    }
}

```

```

        return;
    }

    try {

        double price =
            Double.parseDouble(priceField.getText());

        int quantity =
            Integer.parseInt(quantityField.getText());

        model.addRow(new Object[]{
            idField.getText(),
            nameField.getText(),
            price,
            quantity
        });

        JOptionPane.showMessageDialog(
            this,
            "Product Added Successfully!"
        );

        clearFields();

    } catch (NumberFormatException ex) {

        JOptionPane.showMessageDialog(
            this,
            "Price and Quantity must be numbers."
        );
    }
}

// ----- UPDATE -----

```



```

void updateProduct() {

    int row = table.getSelectedRow();

    if (row == -1) {

        JOptionPane.showMessageDialog(
            this,
            "Select a product to update."
        );

        return;
    }

    int modelRow =
        table.convertRowIndexToModel(row);

    try {

        model.setValueAt(idField.getText(),
            modelRow, 0);

        model.setValueAt(nameField.getText(),
            modelRow, 1);

        model.setValueAt(
            Double.parseDouble(priceField.getText()),
            modelRow, 2);

        model.setValueAt(
            Integer.parseInt(quantityField.getText()),
            modelRow, 3);

        JOptionPane.showMessageDialog(
            this,

```

```

        "Product Updated Successfully!"
    );

    clearFields();

} catch (NumberFormatException ex) {

    JOptionPane.showMessageDialog(
        this,
        "Enter valid price and quantity."
    );
}
}

// ----- DELETE -----

void deleteProduct() {

    int row = table.getSelectedRow();

    if (row == -1) {

        JOptionPane.showMessageDialog(
            this,
            "Select a product to delete."
        );

        return;
    }

    int choice = JOptionPane.showConfirmDialog(
        this,
        "Are you sure you want to delete?",
        "Delete Product",
        JOptionPane.YES_NO_OPTION
    );

```

```

if (choice == JOptionPane.YES_OPTION) {

    int modelRow =
        table.convertRowIndexToModel(row);

    model.removeRow(modelRow);

    JOptionPane.showMessageDialog(
        this,
        "Product Deleted Successfully!"
    );

    clearFields();
}
}

// ----- CLEAR -----

void clearFields() {

    idField.setText("");
    nameField.setText("");
    priceField.setText("");
    quantityField.setText("");

    table.clearSelection();
}

// ----- MAIN -----

public static void main(String[] args) {

    SwingUtilities.invokeLater(() -> {
        new InventoryManagement();
    });
}

```

}

}

Output:

Product

Product Details

Product ID:

1010

Product Name:

Laptop

P101	Laptop	200000.0	
P102	Mouse	800	25
P103	Keyboard	1	
1010	Laptop	200000.0	1

Add

Update

Delete

Clear

Product

Product Details

Product ID:

Product Name:

Price:

Quantity:

Add

Update

Delete

Clear